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RESEARCH ARTICLE

An advanced hybrid quantum tabu search approach to vehicle routing problems

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Abstract
Quantum computing (QC) is expected to solve incredibly difficult problems, including finding optimal solutions to combinatorial optimization problems. However, to date, QC alone has still been unable to demonstrate this capability except for small-sized problems. Hybrid approaches, where QC and classical computing work together, have shown the most potential for solving real-world-scale problems. This work aims to show that we can enhance a classical optimization algorithm with QC so that it can overcome this limitation. We present a new Hybrid Quantum-classical Tabu Search (HQTS) algorithm to solve the Capacitated Vehicle Routing Problem (CVRP). Based on our prior work, HQTS leverages QC for routing within a classical tabu search framework. The quantum component formulates the Traveling Salesman Problem (TSP) for each route as a Quadratic Unconstrained Binary Optimization (QUBO) and solves it using D-Wave's Advantage system. Experiments investigate the impact of quantum routing frequency and starting solution methods. Across different starting solution methods, including quantum-based and classical heuristics, it shows minimal overall impact. HQTS achieved optimal or near-optimal solutions for several CVRP problems, outperforming other hybrid CVRP algorithms and significantly reducing the optimality gap relative to preliminary research. The experimental results demonstrate that more frequent quantum routing improves solution quality and runtime. The findings highlight the potential of integrating QC within meta-heuristic frameworks for complex optimization in vehicle routing problems.

Keywords Quantum optimization · Hybrid quantum/classical algorithms · Meta-heuristic · Vehicle routing · Tabu search

1 Introduction

The current era of Quantum Computing (QC) has brought significant attention. Even outside the immediately related fields, many industries are interested in QC. Industries where quantum computing could create significant disruption if a quantum advantage is achievable, and transportation is one of them. Many transportation industry problems are among the hardest known to solve, including NP-hard problems such as the Traveling Salesman Problem (TSP) and the Vehicle Routing Problem (VRP). QC claims it can solve problems like these more effectively and faster than classical approaches (Abbas et al. 2023).

Although TSPs and VRPs have been extensively studied for many years, they remain an interesting problem due to their direct connection to the transportation industry. Transportation companies like J.B. Hunt Inc. must address these issues daily as part of their standard operations, and QC research is already focusing on these problems (Osaba et al. 2022). In simple terms, the TSP aims to find a tour or route for a salesperson to visit a distinct set of locations while traveling the shortest distance possible. The main difference between the TSP and the VRP is that the TSP produces a single route to visit all locations, whereas the VRP allows multiple routes. These different routes are represented as distinct vehicles. There are many ways to find reasonable solutions to these problems, and

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achieving near-optimality is still valuable for many industry use cases. However, in this work, we aim to determine whether optimal solutions can be found using a hybrid QC algorithm.

This era of QC is most commonly known as the Noisy Intermediate-Scale Quantum (NISQ) era. It is called noisy because the quantum computers of this era produce high error rates. Intermediate-Scale means these same computers do not yet contain many qubits. Most research around solving industry problems relies on toy versions, so the limited number of qubits is sufficient to map the complete problem onto quantum computers. A full CVRP formulation would demand $O(n^2)$ variables, exceeding 10,000 qubits for a problem with just one hundred locations. Other research points to a different direction, where hybrid quantum/classical algorithms are used so that the problems can scale larger. Hybrid algorithms allow portions of the problem to be solved classically, while other portions can be solved using the quantum computer. It can be done so that only some of a given problem needs to be mapped at any given time, reducing the number of qubits required for the quantum computation.

Our Contributions in this Work In this article, we explore ways to enhance a hybrid quantum/classical algorithm by strengthening its quantum components. The proposed algorithm will be further extended from our preliminary work (Holliday et al. 2024), in which we introduced a hybrid algorithm to solve the CVRP. The main refinements over the previous version of the method include alternative starting solutions and the routing frequency. Our refined algorithm significantly improves the preliminary results and outperforms other hybrid quantum algorithms, achieving a lower overall deviation from optimal solutions. Notably, optimal and near-optimal solutions are found for several problems, with one case only 0.03% from the optimum. To achieve these results, we investigate the impact of varying the frequency of quantum routing within a tabu search (TS) algorithm. New results show that more frequent QC use leads to faster convergence to better solutions. We also examine the effect of different starting solutions, including classical heuristics, clustering techniques, and quantum-generated solutions. While results suggest that simpler classical heuristics can be effective starting points, our study does not conclusively demonstrate superiority over quantum-generated solutions. We provide runtime data for the algorithm on various problem sizes, offering insights into the computational cost of the hybrid approach. We highlight the limitations of other hybrid approaches to the VRP, particularly when applied to realistic datasets where optimal routes do not conform to simple cluster structures. Our proposed algorithm outperforms existing approaches in these scenarios.

The remainder of this article is organized as follows. Section 2 will review the background related to this research, including formal definitions. Section 3 introduces our algorithm and explains how it was enhanced based on our preliminary research. Section 4 will focus on our experimental setup and results. Finally, Section 5 will contain the conclusions.

2 Background

In this section, we provide definitions, including the capacitated vehicle routing problem and adiabatic quantum computing. We then discuss methods for solving optimization problems both classically and using QC. Lastly, we explore hybrid approaches to solving the CVRP that use both classical and QC methods.

2.1 Definitions

2.1.1 Vehicle Routing Problem (VRP)

The VRP is formally defined as an NP-hard combinatorial optimization problem with a directed graph $G = (V, E)$, where $V = \{0, 1, ..., n\}$ is the set of locations and $E = \{(v_i, v_j) : v_i, v_j \in V, i \neq j)\}$ is the set of edges between the locations. Location v_0 is not a delivery location; it is the depot where routes begin and end. The set or fleet of vehicles is K. We will discuss constraints around vehicles in the next section. There is also a cost matrix over the edges $C = (c_{ij})_{n\times n}$. The cost is defined as the distance between two locations in the graph.

To solve the VRP, we must create at most $|K|$ routes that start and end at the depot and minimize total cost. For simplicity, we will say the set N equals V, except that it excludes the depot, v_0. x_{ijk} is defined as the binary variable x for an edge between location i and location j on vehicle k. Let u_{ik} be an integer variable representing the visit position of location i on vehicle $k's$ route. We can mathematically represent this problem as a minimization problem with the constraints as follows,

$$min \sum_{k\in K} \sum_{i\in V} \sum_{j\in V} c_{ij} x_{ijk} \tag{1}$$

s. t.

$$\sum_{k\in K} \sum_{j\in V} x_{ijk} = 1 \quad \forall i \in V \tag{2}$$

$$\sum_{j \in N} x_{0jk} = 1 \quad \forall k \in K \tag{3}$$

$$\sum_{i \in N} x_{i0k} = 1 \quad \forall k \in K \tag{4}$$

$$\sum_{i \in N} x_{ijk} - \sum_{i \in N} x_{jik} = 0 \quad \forall j \in N, k \in K \tag{5}$$

$$2 \leq u_{ik} \leq N \quad \forall i \in N, k \in K \tag{6}$$

$$u_{ik} - u_{jk} + 1 \leq (N-1)(1 - x_{ijk}) \quad \forall i, j \in N, k \in K \tag{7}$$

$$x_{ijk} \in \{0, 1\} \tag{8}$$

In this equation, Eq. 1 is the minimization function subject to additional constraints. Constraint Eq. 2 ensures that each location is visited by only one vehicle. Constraints Eqs. 3 and 4 ensure that each route on each vehicle starts and ends at the depot. Constraint Eq. 5 ensures that the number of entries into a location equals the number of exits from it. Constraints Eqs. 6 and 7 prevent sub-tours using the Miller-Tucker-Zemlin formulation (Miller et al. 1960), ensuring routes are connected to the depot without disconnected cycles. Constraint Eq. 8 defines the primary binary decision variable.

2.1.2 Variants of the vehicle routing problem

The CVRP is the VRP variant we focus on in this research. The CVRP introduces a constraint, such that each vehicle's assigned route must adhere to a capacity constraint Q. Therefore, the sum of each location on the route's demand q must be less than or equal to the vehicle's value for Q. Homogeneous and heterogeneous fleet variations exist for a fixed value of Q or a dynamic per-vehicle value of Q. In this research, Q is a fixed value for each vehicle.

$$\sum_{i \in V} q_i \sum_{j \in V} x_{ijk} \leq Q \quad \forall k \in K \tag{9}$$

The constraint in Eq. 9 limits the customers on each vehicle k so that the capacity of the vehicle is not exceeded. This constraint is added to the list of constraints from the previous section to fully define the CVRP.

2.1.3 Adiabatic quantum computing

Today, we can distinguish two types of real quantum devices, including gate-based quantum systems and quantum annealers. On the one hand, a gate-based computer uses qubits to perform simple quantum-circuit operations, analogous to classical operations on bits. It may be joined in any order to form algorithms. A common term for this form is *universal quantum computer*.

On the other hand, a quantum annealer operates on the principle of adiabatic computation, where an initially simple Hamiltonian is gradually evolved from its ground state to the ground state of a final, problem-specific Hamiltonian. If the Hamiltonian evolves slowly enough, the adiabatic theorem ensures that the system stays in the ground state throughout the entire computation. A quantum annealing process starts by representing the problem as an energy landscape. At the outset, the quantum system is initialized in a superposition of all possible states, corresponding to a high-energy state. As the system evolves, the Hamiltonian is modified, enabling it to navigate the energy landscape. The objective is to steer the system toward the lowest-energy state, representing the optimal solution to the problem.

In quantum annealers, the adiabatic theorem is deliberately relaxed, allowing the system to evolve faster than the adiabatic limit typically permits. Consequently, transitions to high-energy states often occur during evolution. To address this situation, innovative methods for reaching adiabaticity have been introduced in the literature (Takahashi 2017; Ferreiro-Vélez et al. 2024).

Finally, although this computational model is also universal (Mizel et al. 2007), the D-Wave quantum annealer employed in this work relies on an Ising Hamiltonian, which limits the types of problems that can be solved on the device. Nevertheless, this type of device is particularly suited for solving combinatorial optimization problems (Yang et al. 2023).

2.2 Classical optimization

Route planning is a prominent topic in artificial intelligence due to its significant scientific and social implications. First, these problems are of great scientific interest because they often involve high computational complexity. As NP-hard problems, they pose a significant challenge for the scientific community. Secondly, routing problems are typically designed to address real-world logistics and transportation scenarios, making their efficient resolution beneficial both socially and commercially.

Numerous methods have been proposed in the literature to tackle these problems efficiently. The most successful methods include exact, heuristic, and meta-heuristic methods (Salhi and Thompson 2022). Among them, metaheuristics are arguably the most popular methods (Hussain et al. 2019). Meta-heuristics explore the solution space to obtain effective optimization results without depending on the specifics of the problem. It makes them especially well-suited to addressing real-world problems with complex formulations,

as they do not require detailed, problem-specific information to explore the feasible solution space.

Furthermore, metaheuristics can be categorized into search-based and constructive algorithms. Search-based techniques begin with an initial complete solution or a set of complete solutions, then are modified until a final solution is reached. In contrast, constructive algorithms begin with a partial solution or a set of partial solutions and incrementally build them until a complete solution is achieved.

The literature features a wide variety of metaheuristics. As the primary aim of this research is not to provide an in-depth description of these algorithms, we recommend that readers refer to Campelo and Aranha (2025) for a comprehensive list of over 100 metaheuristic approaches.

2.3 Quantum optimization

QC represents a groundbreaking advancement in computing, leveraging principles from quantum physics to process information in entirely new ways. Currently, the field is highly anticipated for its potential to address problems that classical computers find intractable, particularly in cryptography, drug discovery, and optimization. This paper focuses explicitly on optimization.

Today, several approaches are being investigated in quantum optimization, with quantum annealing (QA) (Morita and Nishimori 2008) and variational quantum algorithms, such as the quantum approximate optimization algorithm (QAOA) (Farhi et al. 2014), being among the most notable. Despite significant progress, quantum computers remain in their early stages relative to classical computers. They face challenges in solving problems efficiently due to their limited capacity and inherent instability. As a result, we are in the *noisy intermediate-scale quantum* (NISQ) era (Preskill 2018), characterized by the limitations of these devices in effectively handling complex problems.

Despite these challenges, the number of recent studies has increased over the past few years, focusing on solving real-world problems using QC. This increase in publications highlights the community's growing interest in exploring the applications of quantum devices. Several factors have contributed to this intriguing development. Two of these factors have likely had the most influence, including the advancements in QC democratization (Seskir et al. 2023) and the development of increasingly larger and better-connected devices. A primary example of the latter is D-Wave Systems' `Advantage_System`, which comprises 5,616 qubits arranged in a Pegasus topology (Boothby et al. 2020). This system is currently the most widely used for solving optimization problems.

Turning our attention to the topic addressed in this paper, research focused on routing problems using QC has been highly prolific, with academic problems such as the Vehicle Routing Problem (VRP) (Golden et al. 2008) and the Traveling Salesman Problem (TSP) Matai et al. (2010) being the most extensively studied cases. Notably, the survey by Osaba et al. (2022) highlights that this new paradigm has inspired 53 research publications as of 2022. According to the authors of that study, "*it is noticeable that the TSP engages most of the researchers (60,37% - 32 out of 53 papers), while the VRP amounts to 25,52% of the contributions (13 out of 53). The rest of the papers deal with other routing problems, such as the Shortest Path Problem or the Hamiltonian Cycle*". The trend continues, with VRP and TSP remaining the predominant focus of scientific research in the years since 2022. These problems account for the majority of publications, as evidenced by studies such as Le et al. (2023); Spyridis et al. (2023); Qian et al. (2023); Leonidas et al. (2023); Mohanty et al. (2023); Sinno et al. (2023); Tambunan et al. (2023).

Analyzing the current body of work reveals two conclusions. First, many studies aim to assess the potential of quantum technologies or evaluate the efficiency of specific methods, often using academic problems such as the TSP or VRP for benchmarking. Second, numerous studies explore the application of QC to real-world routing problems. The latter research category seeks to maximize the capabilities of current NISQ-era devices by implementing efficient, advanced hybrid-resolution methods. Representative examples of this trend can be found in Osaba et al. (2024); Weinberg (2023).

2.4 Quantum/classical hybrid algorithms for the VRP

As of today, the most recognized article in this field is Feld et al. (2019). They focused on solving the CVRP with a two-phase approach. They experimented with quantum in both phases, but achieved the best results when using it only in the second phase. They presented results on the same dataset we used in our experiments, and we compare their results with ours in Section 4.

As Feld et al. (2019) is an inspirational work, we will describe their algorithm in detail. In the first assignment phase, they implemented a classical clustering algorithm in which each cluster represents a vehicle route. They demonstrated two methods for selecting the starting location for a cluster core, either a location with very high demand or one far from the depot. A cluster is built from the starting location by adding the closest location to the cluster's geometric center, then recalculating the center after the location is added. They continue adding the nearest locations to the geometric center until vehicle capacity is exceeded. They then repeat this process with a new starting location until all locations are clustered. They then perform cluster improvement where they move locations between clusters if the distance from a

location to a cluster center can be decreased by moving it to a different cluster and the capacity of the vehicle is not exceeded. This concludes the first phase of the algorithm.

The second phase, or routing phase, of the heuristic involves formulating the clusters as a QUBO, allowing the TSP to be solved for each one. The TSP QUBO is formulated as defined in Lucas (2014). We present the mathematical formulation of the QUBO because we use the same QUBO in our algorithm. Here we alter the notation from Lucas (2014) and continue with i, j defined as before as the edge from location i to location j, and u is the position in the visit sequence on the route. Thus $x_{i,u}$ is the binary variable indicating location i is visited at position u. N contains all the locations being routed and n is the number of locations in N. E is defined as before as the set of edges between the locations.

$$\begin{aligned} &H_A = \\ &A\sum_{j=1}^{N}\left(1-\sum_{u=1}^{n}x_{j,u}\right)^2 + \\ &A\sum_{u=1}^{N}\left(1-\sum_{j=1}^{n}x_{j,u}\right)^2 + \\ &A\sum_{(i,j)\notin E}\sum_{u=1}^{n}x_{i,u}x_{j,u+1} \end{aligned} \tag{10}$$

Here H_A in Eq. 10 is the QUBO formulation for the Hamiltonian Cycle Problem. The first term ensures that every location appears in the cycle. The second term ensures a *u*th node in the cycle for each *u*. The third term ensures that an edge must exist from *i* to *j*. The summation over $(i, j) \notin E$ denotes pairs (i, j) not in E.

$$H_B = B\sum_{(i,j)\in E} c_{ij}\sum_{u=1}^{n} x_{i,u}x_{j,u+1} \tag{11}$$

H_B in Eq. 11 ensures the cost of the Hamiltonian Cycle is minimized. c_{ij} again is the cost to travel from location i to location j.

$$H = H_A + H_B \tag{12}$$

Adding H_A and H_B together in Eq. 12 provides the complete Hamiltonian H, which is the QUBO that will solve the TSP, or what is commonly referred to as the routing phase of the problem. The penalty coefficients. More recent QC approaches to solving the CVRP include Palackal et al. (2023) and Xie et al. (2024), which use hybrid quantum-classical algorithms, QAOA, and the variational quantum eigensolver (VQE) orithms. The two most performative algorithms were the DBScan Solver (DBSS) and the Solution Partitioning Solver (SPS). DBSS operates in a manner very similar to Feld et al. (2019). SPS can use DBSS as an initial approach, creating a single TSP solution for all locations, and then dividing the TSP into routes to solve the CVRP. They provided their source code, allowing us to recreate their results in our preliminary research. We also experimented with DBSS and SPS to generate the starting solution for our algorithm. It is explained in Section 3.

More recent QC approaches to solving the CVRP include Palackal et al. (2023) and Xie et al. (2024), which use hybrid quantum-classical algorithms, QAOA, and the variational quantum eigensolver (VQE) to solve the problem. Palackal et al. (2023) struggled with QAOA, as it failed to find feasible solutions to toy CVRP problems; however, VQE was successful. Xie et al. (2024) added constraint-preserving mixers to their QAOA and found feasible solutions, but only on toy problems. Suen et al. (2022) implemented a full CVRP QUBO formulation that they solved on the quantum-inspired Fujitsu Digital Annealer (Nakayama et al. 2021). They demonstrated results on CVRPs with up to 50 locations and achieved an optimality gap of 3.89% on a 34-location problem. Sales et al. (2023) presented the most promising recent result, employing a hybrid approach that combines two phases: quantum clustering and classical routing. They also presented results on some of the problems from the same dataset used in our experiments. The result we highlighted was for problem CMT 5, where they outperformed all other hybrid algorithms with an optimality gap of 4.12%. Otherwise, their results were inferior to ours on all other problems. However, their experiments demonstrate the potential of quantum clustering in a two-phase approach, the opposite of the two-phase approach used in Feld et al. (2019).

2.5 Reported advantages of a quantum-annealing subroutine in hybrid optimization

Prior work reports several advantages of QA that are specifically relevant when QA is used as a subroutine within a classical metaheuristic rather than as a stand-alone exact solver. First, quantum annealers can sample ground-state manifolds and near-optimal states in a manner distinct from thermal optimizers; this has been demonstrated experimentally and through comparative analysis of the resulting distributions, suggesting that QA can act as a complementary generator of diverse, high-quality candidates (Zhang 2017). Second, QA has been shown to improve hybrid pipelines by rapidly producing high-quality incumbents that benefit classical solvers on short time scales, providing motivation for its use as a repeated candidate-generation or intensification operator (Pang et al. 2021) Third, a widely used practical pattern is hybrid decomposition in which QA heuristically solves smaller sub-QUBOs while classical local search refines and combines solutions; these approaches explicitly adopt this design and have been analyzed empirically (Boost et al. 2017; Bass et al. 2021) Finally, recent benchmarking

indicates that modern QA/hybrid approaches can be competitive on certain large/dense QUBO families, supporting the broader premise that QA-based subroutines can be useful for particular instance structures when integrated into hybrid algorithms (Kim et al. 2025). Overall, this body of evidence motivates using QA not as a standalone solver, but as a strategically integrated subroutine that leverages its distinctive sampling and incumbent-generation capabilities.

3 Our proposed method

Since NISQ-era quantum optimization is only possible for small-sized problems, we decided to pursue a hybrid approach. We analyzed the experimental results of other hybrid algorithms and found that none achieved optimal performance on the CVRP. All of those hybrid algorithms used a two-phase approach to solve the problem. This approach is discussed in Laporte et al. (2000). We determined that we should explore ways to improve the existing two-phase approaches. We decided that a metaheuristic hybrid approach might be better suited, since our goal is to find optimal solutions.

Metaheuristics have mechanisms to avoid getting stuck in a local optimum. In our approach, we selected TS (Glover and Laguna 1998) as the metaheuristic because it enabled us to incorporate concepts used by other hybrid approaches. Primarily, our approach solves the TSP for each route by utilizing QC. The primary trade-off of using a metaheuristic, in our case, is the algorithm's runtime. The hybrid algorithms we tested, along with the single heuristic, ran much faster but always found suboptimal solutions.

The hybrid design is critical under NISQ limitations to scale our solution to real-world-sized problems. By solving only individual TSP subproblems (≤ 25 nodes), our approach requires ≤ 625 logical qubits per QUBO (Lucas 2014), easily embedded on D-Wave Advantage. Annealing time remains fixed at 20 μs per call, independent of global scale.

3.1 Hybrid Quantum Tabu Search (HQTS)

The research described in Cordeau et al. (2002) states that TS has historically performed well on the CVRP, reinforcing our decision to create a hybrid quantum/classical TS algorithm. TS has been widely studied for solving optimization problems, and various approaches exist for implementing it Taillard (1993), Gendreau et al. (1994), Osman (1993). We can break our algorithm into classical and quantum parts at the highest level. The classical part is our implementation of a TS algorithm. The quantum part is a QUBO formulation of the TSP, solved for each route in a given solution on D-Wave's quantum annealer. We originally named this algorithm Hybrid Quantum Tabu Search. Figure 1 shows a high-level flowchart for HQTS.

3.2 Classical components

TS is a local search optimization technique, with the primary objective of the classical portion of the algorithm being to find the optimal assignment of locations to routes. Starting from a solution, the TS runs in a main loop that performs a local search, making minor changes to the solution and generating a neighborhood of solutions only slightly different from the starting solution. Once the neighborhood is generated, each neighbor solution is evaluated, and the neighbor that yields the largest improvement in the objective function is selected. The chosen solution replaces the starting solution, and the TS returns to the beginning of the main loop, restarting the local search and creating a new neighborhood. It continues until a stopping criterion is met and the best solution found during the search is returned. The name "tabu" is derived from the fact that once a solution is chosen, it is added to a recency memory called the tabu list. While that solution is tabu, meaning it is in the tabu list, it cannot be selected as the next starting solution, except in a specific case we will describe below.

The tabu list keeps track of all recently selected solutions. In our case, the tabu list is treated as short-term memory because only recently selected solutions remain on it, and in our implementation, they remain for only a small number of main-loop iterations. For our algorithm, we allowed a solution to remain in the tabu list for a random number of iterations, drawn from the range $0.4N$ to $0.6N$, where N is the number of locations in the problem. This idea of a random time duration originated in Taillard (1993). The tabu list prevents the TS from behaving like a greedy search algorithm. It now forces the TS to make suboptimal choices at multiple points in the search, thereby driving it into new regions of the global solution landscape.

Local search is the most time-consuming part of the algorithm because it must generate solutions and then select one of them. The neighborhood of solutions is constructed by performing either a (1,0), (0,1), or (1,1) change to the current starting solution. In our method, we define a (0,1) change as swapping two locations on the same route. We also define the (1,0) change as moving a location from one route to another. However, the location sequence in the new route is determined by evaluating which position in the sequence yields the lowest-cost route, as used in Taillard (1993). A (1, 1) change is swapping two locations from two different routes. Once all possible solutions have been generated by applying all possible changes, each is evaluated by computing the total cost of its neighbors. As described earlier, the selected solution from the neighborhood is used to generate a new starting solution, and local search is restarted.

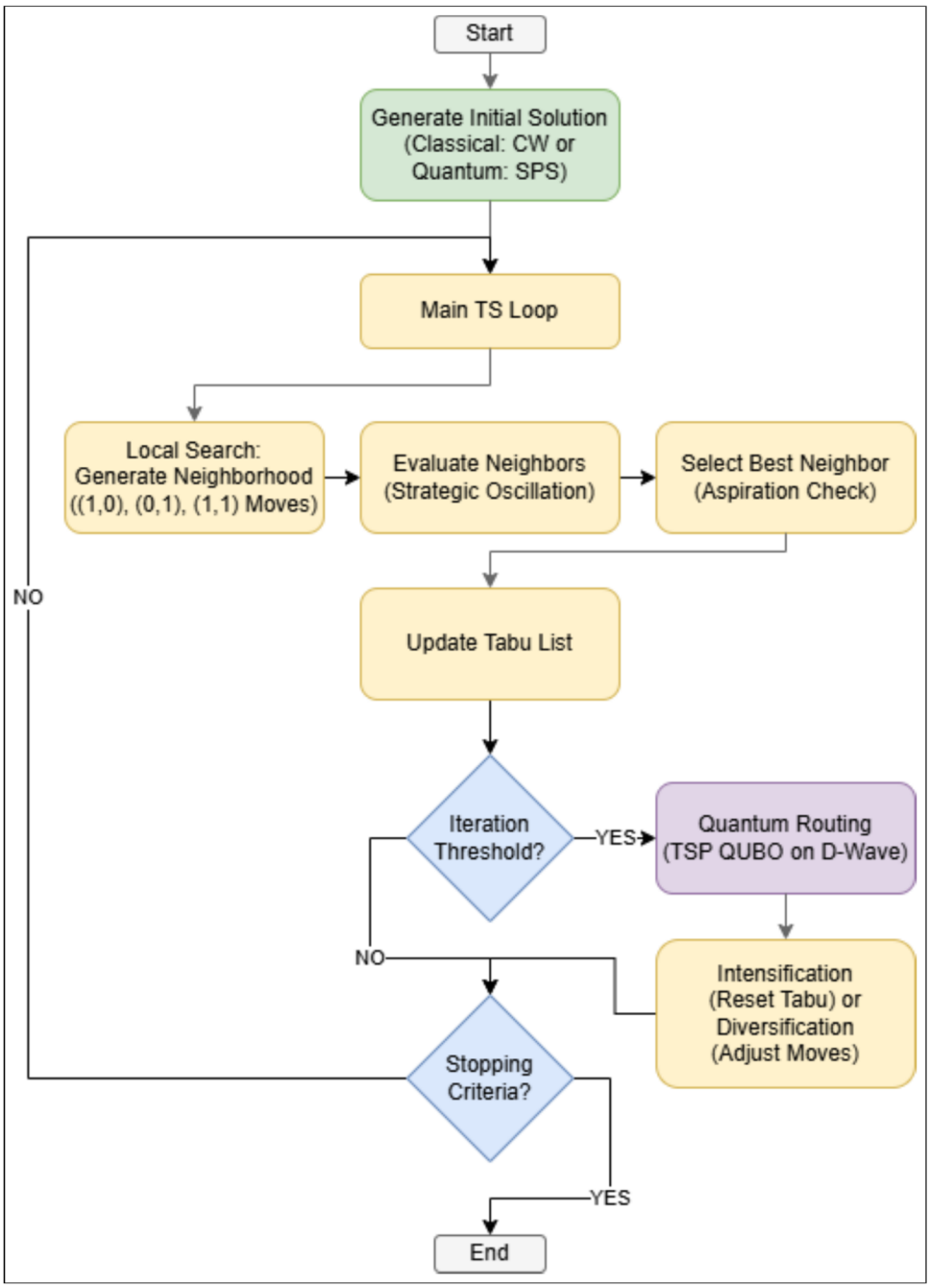


Fig. 1 The proposed flow of HQTS

A condition can occur that allows a chosen solution to be selected even if it already exists in the tabu list. This concept is called aspiration (Glover and Laguna 1998). We define aspiration as the point at which the local search discovers a solution that yields the best value to date for the objective function. It then makes the solution the next starting solution, even if it is in the tabu list. The search does not discard a potentially global optimum solution.

We also implemented other classical TS components. Intensification and diversification are important components (Glover and Laguna 1998). Intensification is the idea of allowing TS to revisit solutions that can search around them more thoroughly. Diversification enables TS to explore previously unseen solutions in the hope of discovering new areas to search. When X main-loop iterations occur without finding a new global-best solution, we trigger intensification or diversification. X was selected uniformly between 0.6 and 1.1 times the number of locations in the problem. These values were determined by experimentation. A diversification trigger changes the number of locations considered when generating new solutions. An intensification trigger resets the tabu list to empty. We also extended our

intensification to include a quantum component, as defined in the next section.

Another interesting TS concept we implemented is strategic oscillation (OS) (Glover and Hao 2011). It is added because Glover and Hao (2011) showed that forcing TS to search only within feasible solutions can limit its effectiveness. Without OS, the algorithm chooses the best feasible solution, with OS, if the initial solution is feasible, the algorithm can select solutions that lead to infeasibility. Once in infeasible territory, the algorithm prioritizes solutions that reduce infeasibility (or minimize its increase) until a feasible solution is found again. This process allows the exploration of a wider solution space. Alg. 1 shows how the next starting solution is selected.

Algorithm 1 Solution evaluation with strategic oscillation.

Require: *cbs* can only be assigned a feasible solution so that it can be evaluated as a global best solution
$Infeasibilty(N_i)$ returns the total capacity violation for a given candidate solution
$p \leftarrow$ previously selected solution
$N \leftarrow$ candidate solutions from local search
$n \leftarrow$ number of candidate solutions
$cbs \leftarrow unassigned$ ▷ current best solution
$sbfs \leftarrow unassigned$ ▷ selected best feasible solution
$sbis \leftarrow unassigned$ ▷ selected best infeasible solution
$ss \leftarrow unassigned$ ▷ selected solution
if $IsFeasible(p)$ **then**
  for $i \leftarrow 1$ to n **do**
    if $Cost(N_i) < Cost(sbfs)$ and $IsFeasible(N_i)$ **then**
      if $Cost(N_i) < Cost(cbs)$ **then**
        $cbs \leftarrow N_i$
      end if
      if $!IsTabu(N_i)$ **then**
        $sbfs \leftarrow N_i$
      end if
    else if $Cost(N_i) < Cost(sbis)$ and $!IsFeasible(N_i)$ **then**
      if $!IsTabu(N_i)$ **then**
        $sbis \leftarrow N_i$
      end if
    end if
  end for
  if $Cost(sbis) < Cost(sbfs)$ **then**
    $ss \leftarrow sbis$
  else
    $ss \leftarrow sbfs$
  end if
else
  for $i \leftarrow 1$ to n **do**
    if $Infeasibilty(N_i) < Infeasibilty(sbfs)$ and $IsFeasible(N_i)$ **then**
      if $Cost(N_i) < Cost(cbs)$ **then**
        $cbs \leftarrow N_i$
      end if
      if $!IsTabu(N_i)$ **then**
        $sbfs \leftarrow N_i$
      end if
    else if $Infeasibilty(N_i) < Infeasibilty(sbis)$ and $!IsFeasible(N_i)$ **then**
      if $!IsTabu(N_i)$ **then**
        $sbis \leftarrow N_i$
      end if
    end if
  end for
  if $Infeasibilty(sbis) < Infeasibilty(sbfs)$ **then**
    $ss \leftarrow sbis$
  else
    $ss \leftarrow sbfs$
  end if
end if

Lastly, we have the stopping criteria. In our preliminary study, we had a hard limit of one hour of wall clock time, or if the main loop had not found a new global best solution in the last 5,000 main loop iterations. In extending this work, we revisited the stopping criteria and allowed longer searches. This was achieved by changing the stopping criterion from stopping after 5,000 main-loop iterations without improving the global best solution to running for $N * 100$ main-loop iterations without improvement, where N is again the number of locations in the problem. We have also removed the one-hour time limit for the wall clock. Section 4 contains details on the run time for our algorithm.

3.3 Quantum components

The quantum components have two different objectives, i.e., one is to initialize the TS with a starting solution to build from, and the other is to determine the best sequence of locations on the routes. When a TS starts, a solution must be provided to perform a local search. In our experiments, we explored various methods for generating the initial solution, including using other quantum hybrid algorithms. In our preliminary research, we created a naive clustering approach to generate the starting solution. We have developed a more advanced system for generating initial solutions. The most famous heuristic for solving the CVRP is the Clarke-Wright savings heuristic (CW) (Clarke and Wright 1964). In our preliminary research, we compared our results with CW and found that CW achieves an average optimality gap of 9.97% on the dataset we tested. That was better than some of the hybrid algorithms we tested. In our preliminary research, we tested hybrid algorithms from Borowski et al. (2020), and the best-performing algorithm achieved an average optimality gap of only 18.68%. Still motivated to incorporate more quantum components into our algorithm, we decided to test two hybrid algorithms and the CW heuristic for generating initial solutions. Section 4 contains the results of our experimentation with these different methods.

The other quantum component in our algorithm was in the intensification process. Following Taillard (1993), we considered calculating the routes (solving the TSP for one route) from the best-known solution at certain points during the search. To do it, we save the best solution found throughout the search. When the quantum routing is triggered, we build a QUBO formulation as described in Eqs. 10, 11 and 12 for the TSP of each route in the best-known solution and then send the QUBO to the quantum computer for optimization. It is done one by one for each route in the solution. The processing time for the quantum computer to solve these TSP QUBO's is in the order of milliseconds. After the quantum computer is finished, we reformat the results in our classical framework. The classical formulation becomes the next starting solution for the next iteration of the main loop. In Taillard (1993), they would perform the routing step every twenty main loop iterations.

Importantly, this quantum component is not intended to outperform state-of-the-art classical TSP solvers in isolation. Rather, it serves as a high-quality stochastic candidate generator for within-route sequencing subproblems within a larger CVRP metaheuristic. This design choice is motivated by several advantages reported in Section 2.5. In our setting, the "subproblem" structure arises naturally (each route defines a small TSP-QUBO), so QA is used as a targeted intensification operator within the classical search, while the full CVRP remains handled classically at scale. Currently, access to QC is not free and involves cost, so the fewer calls to QC, the lower the cost for our algorithm. Additionally, QC is accessed via a cloud service, so more calls increase the risk of an access error. Our preliminary algorithm would trigger this process if TS had not found a new global best solution in the last 2000 iterations of the main loop. In our new approach, we attempted to trigger this form of intensification at varying intervals. Section 4 will show how changing that interval impacted our results.

4 Experiments and results

4.1 Datasets and setup

Extensive research on the VRP and CVRP has produced significant resources, including solutions and benchmarks for evaluating them. For our study, we sought a dataset that would provide a solution the industry could find interesting. A dataset with problems that are representative of real-world routing problems. Typically, when routing vehicles around a metropolitan area, you will likely find a depot centrally located among the points it serves. In this scenario, routes often resemble flower petals: they leave the depot, make deliveries as they arc away, and return to their starting location. In some cases, deliveries will be very clustered, but that is not the rule. With that in mind, there is a dataset well suited to this type of routing evaluation, i.e, the dataset from Christofides, Mingozzi, and Toth (CMT) (Christofides et al. 1979). When visualizing the optimal routes for this dataset, you can often see that flower shape. A visual representation of Problem 1 and a visualization of the optimal solution obtained from HQTS are shown in Figs. 2 and 3. Optimal route visualizations for all of the CMT datasets are available at http://vrp.galgos.inf.puc-rio.br/index.php/en/.

The CMT dataset was chosen for additional reasons. This kind of dataset provides a substantial benchmark to compare against because of its history, so you can immediately see

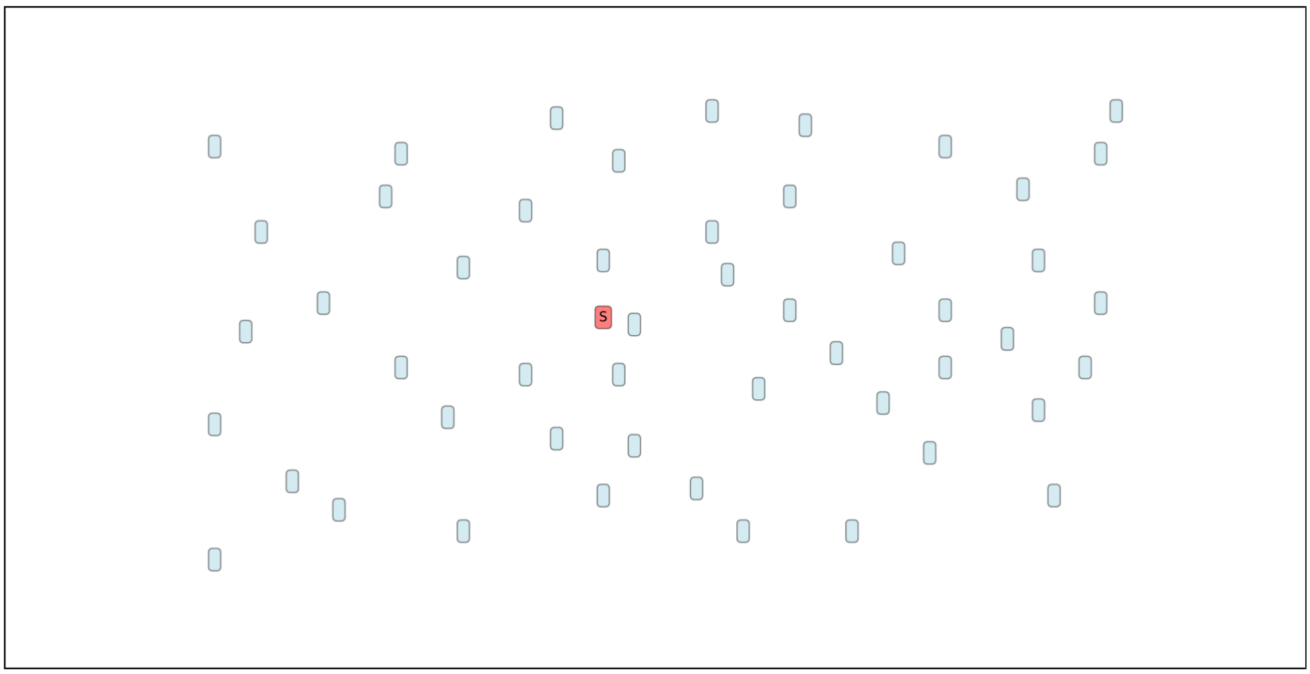


Fig. 2 Visualization of CMT 1 (red = depot, gray = delivery location)

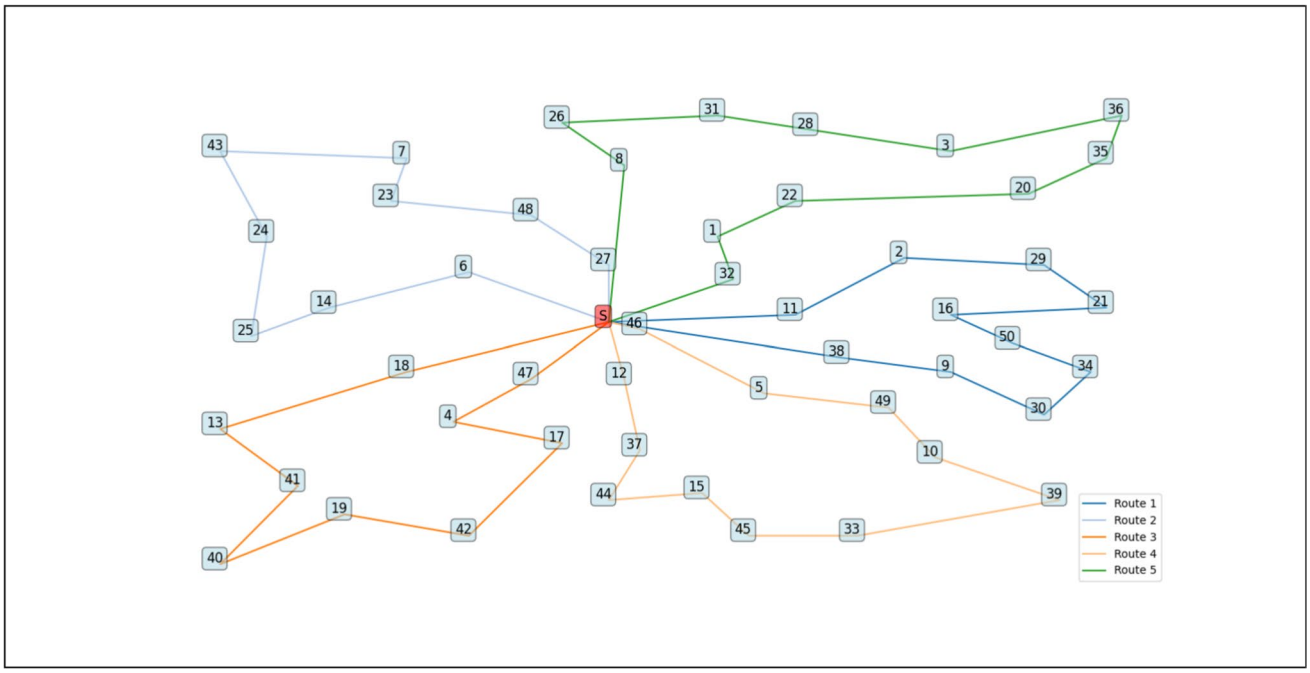


Fig. 3 CMT 1 optimal result using HQTS

how your algorithm ranks with other research. We identified additional hybrid algorithms (Feld et al. 2019; Borowski et al. 2020; Sales and Araos 2023) that have demonstrated performance on this dataset. Another important aspect of this dataset is that the optimal solution to each problem has been found and proven. The dataset consists of 14 problems, with problems 1-5 having varying numbers of locations, ranging from 50 to 199. Table 1 presents the details of each problem. Additionally, the locations are randomly located around a central depot. Problems 11 and 12 have more clustered locations. In 11, the depot is centrally located, whereas in 12, it is shifted to one side. Problems 6-10 follow the same layout as problems 1-5, but include an additional time-window detail for each location. This applies to problems 13 and 14 as

Table 1 CMT dataset

Problem No.	No. of Locations	Location Distribution
1	50	Random
2	75	Random
3	100	Random
4	150	Random
5	199	Random
6	50	Random
7	75	Random
8	100	Random
9	150	Random
10	199	Random
11	120	Cluster
12	100	Cluster
13	120	Cluster
14	100	Cluster

well. We solved only problems 1-5, 11, and 12 because they were explicitly formulated as CVRP problems. The other problems are similar but designed for the CVRPTW, which we did not attempt to solve with our algorithm. Although the locations appear to be randomly placed, they effectively represent a real-world problem.

All of our experiments were run in GitHub Codespaces (Github and Codespaces 2024). The Codespace had a four-core processor with 16 GB of RAM that was used for the classical computation of our algorithm. The quantum part of the algorithm was executed on D-Wave's Advantage System 4.1. We accessed this system via D-Wave's cloud API (Inc 2024). The `num_reads` parameter was set to 1000 for all calls to the QC.

4.2 Preliminary results examined

In our preliminary work (Holliday et al. 2024), we compared our solution against other hybrid solutions. Specifically, we compared our results with those reported by the hybrid algorithm in Feld et al. (2019) and by the DBSS and SPS algorithms in Borowski et al. (2020), which we re-implemented using the provided code. For this problem, CMT 1, our algorithm found the BKS, which, as stated above, is also the optimal solution to the problem. That is the only problem where we achieved optimality. Still, our algorithm outperformed the other hybrid solutions on four of the seven problems we tested on the CMT dataset, resulting in a lower overall percentage deviation from the optimal solution than the other hybrid algorithms. We also compared our results against several well-known heuristics and outperformed them on three of six problems. The heuristics were Clarke-Wright (Clarke and Wright 1964), Fisher-Jaikumar (Fisher 1981), and Sweep (Gillett and Miller 1974). The heuristic's results were reported in Feld et al. (2019). While one result was encouraging when we found an optimal solution, performance fell short of expectations in that our algorithm was unable to outperform the other hybrid algorithms on each problem.

4.3 Ablation study on HQTS configurations

We compare the refined HQTS against a baseline configuration from Holliday et al. (2024), treating prior results as an earlier parameter setting. In this study, we compared our new results with our preliminary results and with the results reported in Feld et al. (2019), which was the second-highest-performing hybrid algorithm in our preliminary work. This comparison is illustrated in Table 5 and Fig. 7. For reference, we also present a comparison of our new results with those obtained using the same classical heuristics reported in our preliminary research. This comparison is illustrated in Table 6.

For this new research, three main questions drove our experimentation. Could we use more QC in our process, and would doing so improve our results? Could we use QC to generate an initial solution, and how would that compare with a heuristic or clustering approach? Thereafter, in answering the previous two questions, would that help us find better solutions than before?

4.3.1 Question 1: quantum routing delay

To answer the first question, we conducted a test in which we varied the frequency of executing the quantum routing component of the algorithm. In our preliminary research, we chose a delay of 2000 iterations of the search without any improvement before performing the quantum routing. The number 2000 was chosen arbitrarily and is primarily intended to limit our use of the quantum computer, given our very limited QC time. We decided to test different delay values. The experiments ranged from 3,000 to 250 (3,000, 2,500, 2,000, 1,500, 1,000, 500, 250). To experiment, we ran the algorithm on CMT 1. If the result was equal to the BKS or within a 2% deviation from optimal, we considered it a good run. Specifically for CMT 1, the allowed solution range was 524.61 to 532.8. We accumulated 10 good runs for each delay value. We cataloged the average number of iterations to the best solution and the average wall-clock time (run time) in seconds for the ten good runs at each delay value in Table 2.

The results show that the best performance across both metrics was achieved with a quantum routing delay of 250. This means allowing quantum routing to trigger more frequently and more quickly when non-improving iterations occur, enabling the algorithm to find a better solution faster, measured by both the number of iterations and wall-clock time. Figure 4 shows, on average, the decreasing number of

Table 2 Average results for different quantum routing delays on problem CMT 1

Quantum Routing Delay	Iterations To Best Solution Found	Run Time (seconds)
3000	8899	433
2500	7153	411
2000	7510	400
1500	5458	365
1000	4336	322
500	2926	290
250	1811	255

iterations needed to find the best solution, and Fig. 5 shows, on average, the decreasing run time in seconds to find the best solution.

4.3.2 Question 2: starting solution

For the second question, we initially used different starting solutions, with a quantum routing delay of 2000. The first option for the starting solution was the clustered approach used in our preliminary research (see that work for an explanation). The next step was to use the Clarke-Wright savings heuristic (CW) (Clarke and Wright 1964). For the remaining two, we used hybrid quantum approaches. These approaches were introduced in Borowski et al. (2020), and we recreated them in our preliminary research, specifically in the Solution Partitioning Solver with DBSCAN (SPSDBSCAN) and Solution Partitioning Solver without DBSCAN (SPSFULLQ).

SPSFULLQ was the best-performing hybrid approach we could re-create. However, it is limited to running on CMT 1 because, for larger problems, that algorithm produces a QUBO that is too large to solve on the DWAVE quantum annealer. The hybrid solutions are noisy, producing different results each time they are run, while clustered and CW produce the same solution every time. We again used CMT 1 for these tests, with the same requirement of 10 good runs per test. A good run is defined the same as in the previous experiment. Table 3 shows the average results for this experiment. The results here were somewhat surprising in that a more minimal starting solution did not lead to finding a good solution faster.

We then used the results from the quantum routing experiment, changing the delay from 2000 to 250, and repeating the experiment with only SPSDBSCAN and CW. We also extended this experiment into CMT 1, CMT 2, and CMT 3. The results here did not show that either starting solution was consistently faster at finding a better solution. Figure 6 shows that SPSDBSCAN was faster on CMT 1 and CMT 3, but CW was faster on CMT 2. For CMT 2 and CMT 3, we were unable to find the optimal solution. Our best runs used CW, but the results are very close on average, as shown in Table 4.

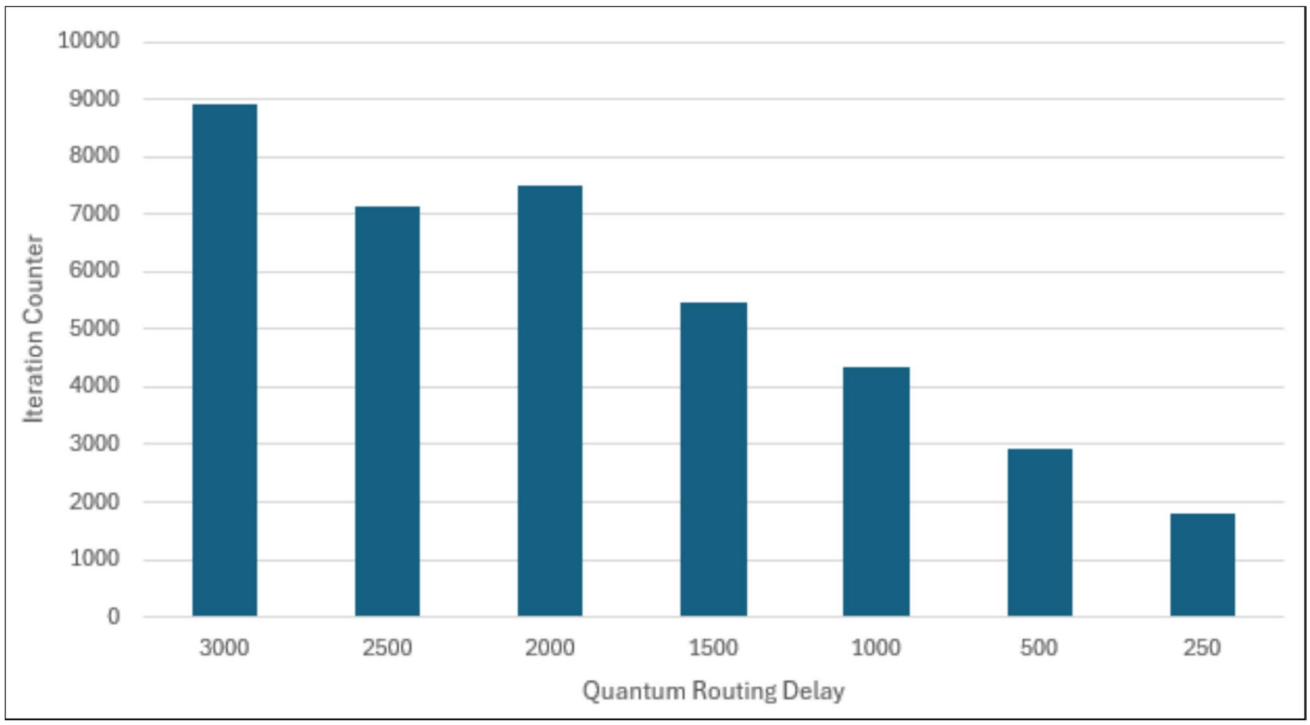


Fig. 4 Average no. of iterations to best solution found with different quantum routing delays on CMT1

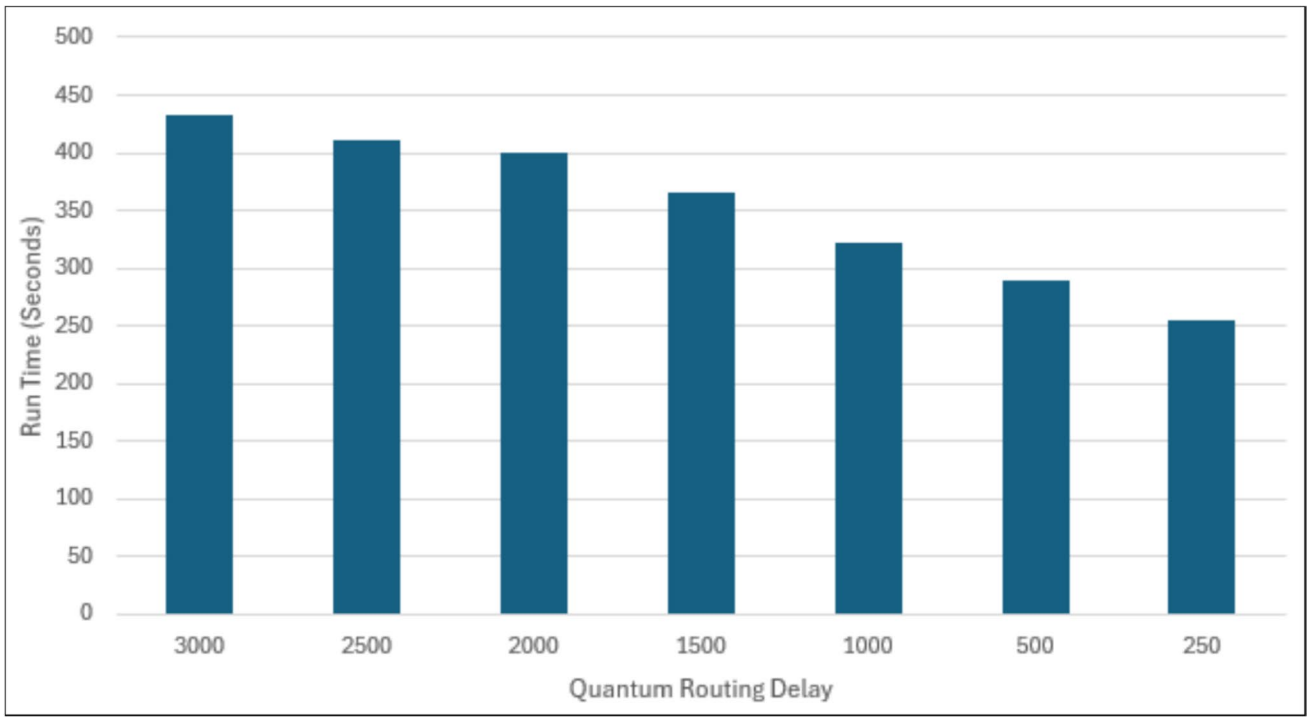


Fig. 5 Average run time to best solution found with different quantum routing delays on CMT1

Table 3 Average performance comparison for different starting solutions on CMT 1 (quantum routing delay 2000)

Algorithm	Starting Solution	Iterations To Best Solution Found	Distance
SPS (DBSCAN)	662.09	**3120**	527.83
SPS (FULLQ)	714.53	5857	**524.61**
Cluster	1108.65	7885	526.44
CW	**584.41**	7510	526.54

4.3.3 Question 3: overall improvement

To answer the third question, we applied the answers to questions one and two to the same experiment we conducted in our preliminary research. We ran the tests as before, retaining the best solution, so our algorithm runs only three times. We then adjusted aspects of the algorithm based on the results generated from questions one and two. We set the routing delay to 250 for each experiment and used the CW heuristic to generate the initial solution. We decided to use CW because we found better solutions with the heuristic, even if only slightly, and it would allow us to use less QC in our tests. We wanted to maximize our available QC resources to complete our experiments, and the results of question two showed little difference in the starting solution for our algorithm.

In Table 5, we match our CMT 1 results and improve our results on all remaining problems. Again, our results show that we found only one optimal solution across the seven problems. However, for problem CMT 12, our new result was 0.03% away from the optimal solution. On the five remaining problems, we documented improved results. We found better solutions for each problem, outperforming the hybrid algorithm of Feld et al. (2019). Overall, we reduced our optimality gap from 4.72% to 2.15%. Refer to Fig. 7. Table 6 also compares these results with those of well-known heuristics on the same problems.

In our preliminary results, we did not report the runtime of the algorithm, so we have decided to include those results as well. Run time was not limited at all in these results. Those runtimes range from 255 seconds for the smallest CMT 1 problem to 37868 seconds for the largest CMT 5 problem. The average run time for each problem is reported in Table 7.

5 Conclusions

We aimed to show that a hybrid quantum algorithm can find optimal solutions to a real-world optimization problem. While our research identified only one optimal solution and nearly one more, we still demonstrated that QC helps find optimal solutions. To date, this is the best-performing hybrid algorithm for the CVRP on the CMT dataset.

We outperformed our previous results on every problem, primarily by using QC more frequently during the experiments. Additionally, using better starting solutions led to

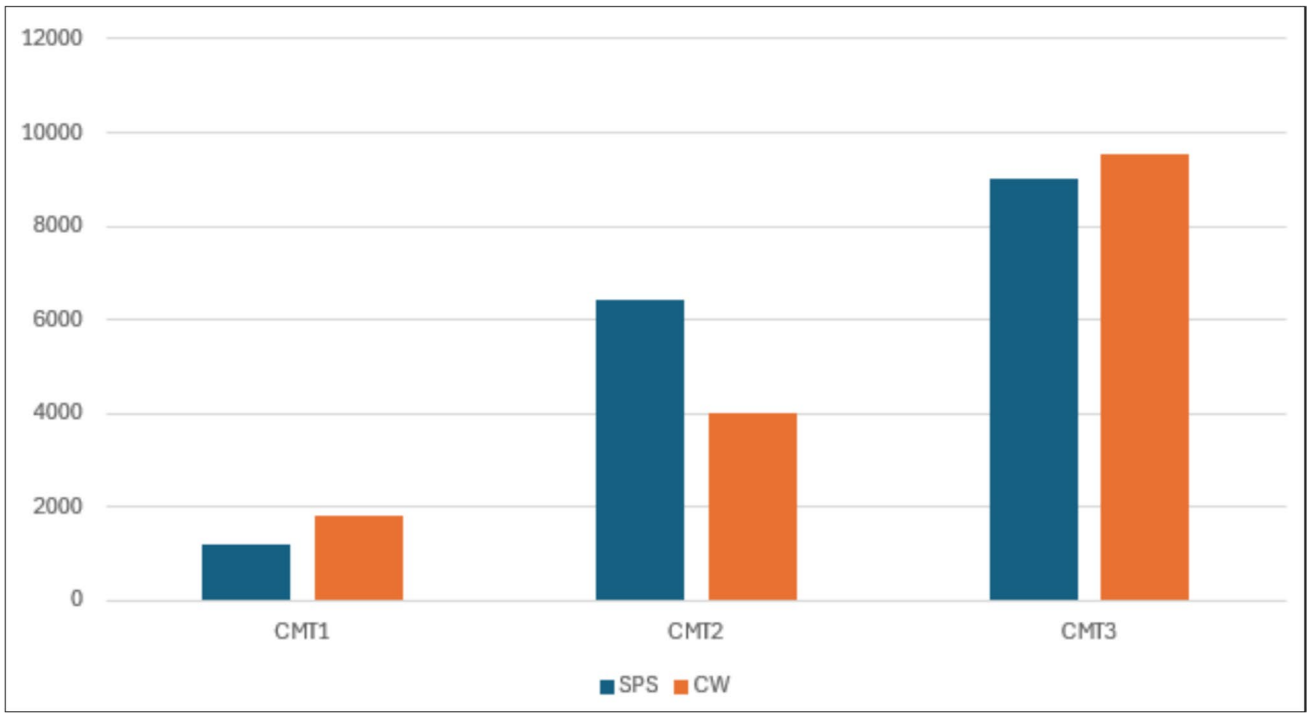


Fig. 6 Average no. of iterations to best solution found with different starting solutions

Table 4 Performance comparison for different starting solutions (quantum routing delay 250)

	SPS (DBSCAN)			CW		
Problem	Avg. Result	Best Result	Avg. Iterations	Avg. Result	Best Result	Avg. Iterations
CMT 1	**525.43**	**524.61**	**1183**	526.54	**524.61**	1811
CMT 2	**851.76**	851.58	6401	855.7	**848.9**	**4018**
CMT 3	**835.36**	831.37	**9020**	836.62	**830.99**	9556

Table 5 Performance comparison of HQTS on the CMT dataset

			Feld at el Ref. (Feld et al. 2019)		Preliminary Ref. (Holliday et al. 2024)		New Research	
Problem	Size	BKS	Distance	Dev.	Distance	Dev.	Distance	Dev.
CMT 1	50	524.61	556	5.98%	**524.61**	**0.0%**	**524.61**	**0.0%**
CMT 2	75	835.26	926	10.86%	856	2.52%	**848.95**	**1.64%**
CMT 3	100	826.14	905	9.55%	876	6.06%	**830.99**	**0.59%**
CMT 4	150	1,028.42	1,148	11.63%	1,094	6.4%	**1,076.56**	**4.68%**
CMT 5	199	1,291.29	1,429	10.66%	1,442	11.72%	**1,359.08**	**5.25%**
CMT 11	120	1,042.12	1,084	4.02%	1,096	5.19%	**1,071.9**	**2.86%**
CMT 12	100	819.56	828	1.03%	829	1.16%	**819.77**	**0.03%**

improved results. However, we could not demonstrate that a quantum-computing-derived starting solution was always superior to a classical-heuristic-algorithm-derived starting solution. Our average runtime was slightly faster, and we found slightly more optimal solutions with the quantum starting solutions. However, using the classical heuristic as the starting solution identified the optimal solutions for CMT 2 and CMT 3. Identifying additional applications for the quantum computer was the primary focus of our research, and our new results showed that using more QC was generally beneficial.

Results from Section 4.3.1 indicate a trend toward improved convergence with reduced routing delay. Although further reductions below 250 iterations may yield additional gains, experimental constraints prevented confirmation. Access to QC resources was governed by a fixed quota that could not be adjusted during execution. Exceeding this quota by invoking the quantum solver too frequently would

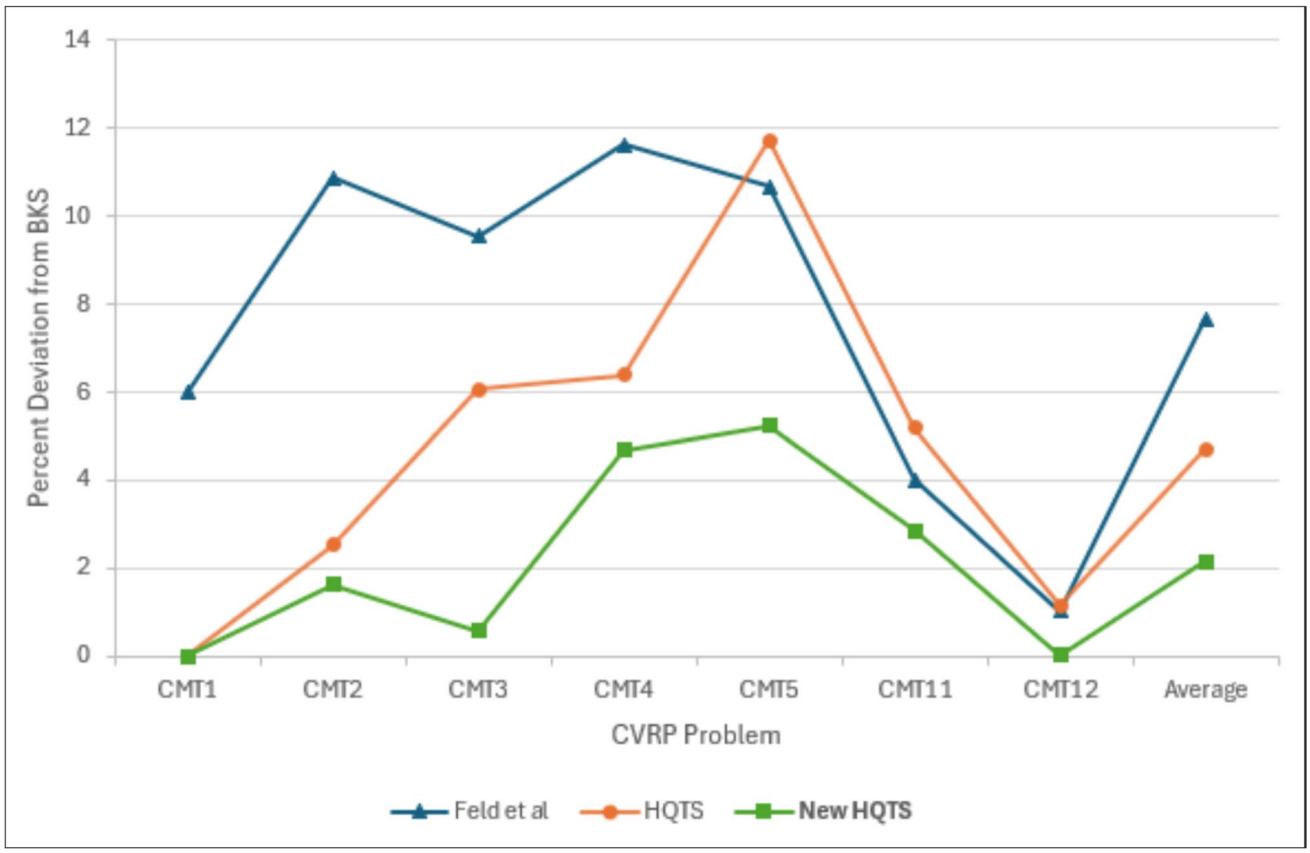


Fig. 7 Percent deviation from BKS on CMT dataset

Table 6 Performance comparison of well-known heuristics and HQTS on the CMT dataset

			Clarke-Wright		Fisher-Jaikumar		Sweep		New Research	
Problem	Size	BKS	Distance	Dev.	Distance	Dev.	Distance	Dev.	Distance	Dev.
CMT 1	50	524.61	585	11.5%	524	0.12%	532	1.41%	**524.61**	**0.0%**
CMT 2	75	835.26	900	7.75%	857	2.6%	874	4.64%	**848.95**	**1.64%**
CMT 3	100	826.14	886	7.25%	833	0.83%	851	3.01%	**830.99**	**0.59%**
CMT 4	150	1,028.42	1,204	17.07%	-	-	1,079	4.92%	**1,076.56**	**4.68%**
CMT 5	199	1,291.29	1,540	19.26%	1,420	9.97%	1,389	7.57%	**1,359.08**	**2.86%**
CMT 12	100	819.56	877	7.01%	848	3.47%	949	15.79%	**819.77**	**0.03%**

Table 7 Average run times on the CMT dataset

Problem	Size	Run Time (Seconds)
CMT 1	50	255
CMT 2	75	1534
CMT 3	100	3807
CMT 4	150	17687
CMT 5	199	37868
CMT 11	120	1886
CMT 12	100	1169

terminate the entire run. Future work will require expanded quantum resource allocation to enable systematic evaluation of delays below 250.

While the two-phase approach to solving the CVRP has been well studied and the compared hybrid quantum algorithms utilized those ideas (Feld et al. 2019; Borowski et al. 2020), our results showed they do not always do well with a dataset like CMT. They show fairly good performance on CMT 11 and CMT 12 problems, which is to be expected given the more clustered locations. They fall behind our results on the randomized CMT 1-5 problems. The shape of the optimal routes was a key design decision in our algorithm. Analysis suggests the clustering approach for route generation does not do a good job of finding the correct clusters to base the routes on. Because the route shape visualization resembles a flower petal, with some stops closer to the depot and others farther away, a clustering algorithm may overlook potentially good locations to add to the route, since they are closer to the depot and not near other locations in the cluster.

The complexity of the large CVRP problems limited the algorithm's ability to find optimal solutions. Analysis indicates that the local search component of the algorithm was the primary reason no more optimal solutions were found. The assignment of locations to routes was more consequential to the suboptimal results than the quantum routing. The neighborhood generation relied exclusively on basic relocation and swap operators. The operations included only two types of moves: relocating a single customer from one route to another, or swapping two customers across different routes. More sophisticated operators, such as those introduced in Lin and Kernighan (1973), were not implemented. Future work will focus on developing a quantum-enhanced local search mechanism to generate higher-quality neighbor solutions with greater efficiency.

Author Contributions J.H and E.O wrote the main manuscript. J.H prepared pseudo code, result tables, figures, and experiment setup. K.L discussed the novelty and the research direction. All the authors revised the manuscript.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing Interests The authors declare no competing interests.

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References

Abbas A, Ambainis A, Augustino B, Bärtschi A, Buhrman H, Coffrin C, Cortiana G, Dunjko V, Egger DJ, Elmegreen BG et al (2023) Quantum optimization: Potential, challenges, and the path forward. arXiv preprint arXiv:2312.02279

Ariño Sales JF, Palacios Araos RA (2023) Adiabatic quantum computing impact on transport optimization in the last-mile scenario. Front Comput Sci 5:1294564

Bass G, Henderson M, Heath J, Dulny J III (2021) Optimizing the optimizer: decomposition techniques for quantum annealing. Quantum Mach Intell 3(1):10

Boost M, Reinhardt S, Roy A (2017) Partitioning optimization problems for hybrid classical/quantum execution. D-Wave Syst., D-Wave Technical Report Series, Burnaby, CO, Canada, Tech. Rep

Boothby K, Bunyk P, Raymond J, Roy A (2020) Next-generation topology of d-wave quantum processors. arXiv preprint arXiv:2003.00133

Borowski M, Gora P, Karnas K, Błajda M, Król K, Matyjasek A, Burczyk D, Szewczyk M, Kutwin M (2020) New hybrid quantum annealing algorithms for solving vehicle routing problem. In: International conference on computational science. Springer, pp 546–561

Campelo F, Aranha C (2025) Evolutionary computation bestiary. https://fcampelo.github.io/EC-Bestiary/. Accessed on 1 Nov 2025

Christofides N, Mingozzi A, Toth P (1979) Combinatorial Optimization. Wiley, Chichester, West Sussex

Clarke G, Wright JW (1964) Scheduling of vehicles from a central depot to a number of delivery points. Oper Res 12(4):568–581

Cordeau J-F, Gendreau M, Laporte G, Potvin J-Y, Semet F (2002) A guide to vehicle routing heuristics. J Oper Res Soc 53:512–522

Farhi E, Goldstone J, Gutmann S (2014) A quantum approximate optimization algorithm. arXiv preprint arXiv:1411.4028

Feld S, Roch C, Gabor T, Seidel C, Neukart F, Galter I, Mauerer W, Linnhoff-Popien C (2019) A hybrid solution method for the capacitated vehicle routing problem using a quantum annealer. Frontiers in ICT 6:13

Ferreiro-Vélez J, Iriarte-Zendoia I, Ban Y, Chen X (2024) Shortcuts for adiabatic and variational algorithms in molecular simulation. https://arxiv.org/abs/2407.20957

Fisher ML, Jaikumar A (1981) A generalized assignment heuristic for vehicle routing. Networks 11(2):109–124

Gendreau M, Hertz A, Laporte G (1994) A tabu search heuristic for the vehicle routing problem. Manage Sci 40(10):1276–1290

Gillett BE, Miller LR (1974) A heuristic algorithm for the vehicle-dispatch problem. Oper Res 22(2):340–349

GitHub I (2024) GitHub codespaces. https://github.com/features/codespaces. Accessed on 02 Oct 2024

Glover F, Hao J-K (2011) The case for strategic oscillation. Ann Oper Res 183:163–173

Glover F, Laguna M (1998) Tabu Search. Springer, New York

Golden BL, Raghavan S, Wasil EA (2008) The Vehicle Routing Problem: Latest Advances and New Challenges. Springer, vol 43

Holliday J.B., Morgan B, Churchill H, Luu K (2024) Hybrid quantum tabu search for solving the vehicle routing problem. In: 2024 IEEE international conference on quantum computing and engineering (QCE). IEEE, pp 353–358

Hussain K, Mohd Salleh MN, Cheng S, Shi Y (2019) Metaheuristic research: a comprehensive survey. Artif Intell Rev 52:2191–2233

Inc D-WS (2024) D-Wave Leap Log In | D-Wave Leap™. Accessed on September 25 2024. https://cloud.dwavesys.com/leap/login/?next=/leap/

Kim S, Ahn S-W, Suh I-S, Dowling AW, Lee E, Luo T (2025) Quantum annealing for combinatorial optimization: a benchmarking study. NPJ Quantum Inf 11(1):77

Laporte G, Gendreau M, Potvin J-Y, Semet F (2000) Classical heuristics for the vehicle routing problem. Int Trans Oper Res 7:285–300. https://doi.org/10.1111/j.1475-3995.2000.tb00200.x

Le TV, Nguyen MV, Khandavilli S, Dinh TN, Nguyen TN (2023) Quantum annealing approach for selective traveling salesman problem. In: ICC 2023-IEEE international conference on communications. IEEE, pp 2686–2691

Leonidas I.D., Dukakis A, Tan B, Angelakis DG (2023) Qubit efficient quantum algorithms for the vehicle routing problem on quantum computers of the nisq era. arXiv preprint arXiv:2306.08507

Lin S, Kernighan BW (1973) An effective heuristic algorithm for the traveling-salesman problem. Oper Res 21(2):498–516

Lucas A (2014) Ising formulations of many np problems. Front Phys 2:74887

Matai R, Singh SP, Mittal ML (2010) Traveling salesman problem: an overview of applications, formulations, and solution approaches. Travel Salesman Probl Theory Appl 1(1):1–25

Miller CE, Tucker AW, Zemlin RA (1960) Integer programming formulation of traveling salesman problems. J ACM (JACM) 7(4):326–329

Mizel A, Lidar DA, Mitchell M (2007) Simple proof of equivalence between adiabatic quantum computation and the circuit model. Phys Rev Lett 99:070502. https://doi.org/10.1103/PhysRevLett.99.070502

Mohanty N, Behera B.K., Ferrie C (2023) Analysis of the vehicle routing problem solved via hybrid quantum algorithms in the presence of noisy channels. IEEE Trans Quantum Eng

Morita S, Nishimori H (2008) Mathematical foundation of quantum annealing. J Math Phys 49(12)

Nakayama H, Koyama J, Yoneoka N, Miyazawa T (2021) Description: Third generation digital annealer technology. Tokyo, Japan, Fujitsu Limited

Osaba E, Villar-Rodriguez E, Asla A (2024) Solving a real-world package delivery routing problem using quantum annealers. Sci Rep 14(1):24791

Osaba E, Villar-Rodriguez E, Oregi I (2022) A systematic literature review of quantum computing for routing problems. IEEE Access 10:55805–55817

Osman IH (1993) Metastrategy simulated annealing and tabu search algorithms for the vehicle routing problem. Ann Oper Res 41:421–451

Palackal L, Poggel B, Wulff M, Ehm H, Lorenz JM, Mendl CB (2023) Quantum-assisted solution paths for the capacitated vehicle routing problem. In: 2023 IEEE international conference on quantum computing and engineering (QCE). IEEE, vol 1, pp 648–658

Pang Y, Coffrin C, Lokhov AY, Vuffray M (2021) The potential of quantum annealing for rapid solution structure identification. Constraints 26(1):1–25

Preskill J (2018) Quantum computing in the nisq era and beyond. Quantum 2:79

Qian W, Basili RA, Eshaghian-Wilner MM, Khokhar A, Luecke G, Vary JP (2023) Comparative study of variations in quantum approximate optimization algorithms for the traveling salesman problem. Entropy 25(8):1238

Sales JFA, Araos RAP (2023) Adiabatic quantum computing for logistic transport optimization. arXiv preprint arXiv:2301.07691

Salhi S, Thompson J (2022) An overview of heuristics and metaheuristics. Palgrave Handbook Oper Res 353–403

Seskir ZC, Umbrello S, Coenen C, Vermaas PE (2023) Democratization of quantum technologies. Quantum Sci Technol 8(2):024005

Sinno S, Groß T, Mott A, Sahoo A, Honnalli D, Thuravakkath S, Bhalgamiya B (2023) Performance of commercial quantum annealing solvers for the capacitated vehicle routing problem. arXiv preprint arXiv:2309.05564

Spyridis Y, Gkelias A, Argyriou V (2023) Variational quantum approach for the multiple traveling salesman problem optimisation. In: 2023 19th International conference on distributed computing in smart systems and the internet of things (DCOSS-IoT). IEEE, pp 354–358

Suen WY, Parizy M, Lau HC (2022) Enhancing a qubo solver via data driven multi-start and its application to vehicle routing problem. In: Proceedings of the genetic and evolutionary computation conference companion, pp 2251–2257

Taillard É (1993) Parallel iterative search methods for vehicle routing problems. Networks 23(8):661–673

Takahashi K (2017) Shortcuts to adiabaticity for quantum annealing. Phys Rev A 95(1). https://doi.org/10.1103/physreva.95.012309

Tambunan TD, Suksmono AB, Edward IJM, Mulyawan R (2023) Quantum annealing for vehicle routing problem with weighted segment. In: AIP conference proceedings, vol 2906. AIP Publishing

Weinberg SJ, Sanches F, Ide T, Kamiya K, Correll R (2023) Supply chain logistics with quantum and classical annealing algorithms. Sci Rep 13(1):4770

Xie N, Lee X, Cai D, Saito Y, Asai N, Lau HC (2024) A feasibility-preserved quantum approximate solver for the capacitated vehicle routing problem. Quantum Inf Process 23(8):291

Yang Z, Zolanvari M, Jain R (2023) A survey of important issues in quantum computing and communications. IEEE Commun Surv Tutor 25(2):1059–1094

Zhang BH, Wagenbreth G, Martin-Mayor V, Hen I (2017) Advantages of unfair quantum ground-state sampling. Sci Rep 7(1):1044

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